SONG QIAOYUCHEN

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EDUCATION

National University of Singapore **Aug 2024 - Present**

Double Degree in Computer Science and Mathematics (GPA: 4.94/5.00)

- Relevant courseworks: Programming Methodology, Computer Organisation, Data Structure and Algorithms, Software Engineering, Database Systems, Computer Networks, Information Security, Design Thinking
- *NUS Overseas College (Kuala Lumpur, Malaysia) Entrepreneurship Programme*
- *NUS College Interdisciplinary Programme*
- NUS Greyhats: Co-organised GreyCTF, Singapore's largest student-run CTF with 1000+ participants

SKILLS

- Languages: English, Mandarin (Native / Bilingual)
- **Programming Languages**: JavaScript, TypeScript, Java, C#, Python, C, C+, Go
- **Frontend & UI**: HTML, CSS, React, Redux, Next.js, ASP.NET, WPF, Material UI, Tailwind CSS, Three.js, R3F
- **Backend & Databases**:SQL (PostgreSQL, Microsoft SQL Server), NoSQL (Neo4j, Supabase), RESTful API, FastAPI
- Design: Clip Studio Paint, Adobe Photoshop, Adobe Illustrator

WORK EXPERIENCE

Sonicboom Solutions, *Software Engineer Intern* **May 2025 - Jul 2025**

- Redeveloped the Hawkeye Console, a WPF-based real-time monitoring system used to track parking gantry status and operational events across 13 deployment sites.
- Implemented MQTT-based messaging to enable reliable real-time communication between parking gantries and the monitoring console, supporting asynchronous event streaming and system alerts.
- Refactored UI components and data-binding logic in WPF to improve maintainability and responsiveness of the monitoring interface.
- Collaborated with system operators to identify usability issues and implemented UI/UX improvements for time-critical monitoring workflows.

PROJECTS

Uniplans, *NUS Course Planning Web App* **May 2025 - Dec 2025**

- Founded and led the development of uniplans.net, a course planning platform that helps NUS students visualise degree requirements and generate informed academic plans under complex prerequisite constraints.
- Optimized prerequisite dependency queries by modeling complex academic relationships in Neo4j, reducing graph traversal time compared to relational lookups.
- Designed and implemented a greedy scheduling algorithm to generate feasible, near-optimal course plans, balancing academic constraints with user preferences such as workload and graduation timeline.
- Built an interactive drag-and-drop planner with real-time client-side validation, optimising frontend performance through normalised state, memoised selectors, and isolating the presentational layer to reduce re-render overhead at scale.
- Applied user-centred design principles to visualise complex prerequisite graphs in an intuitive and accessible interface.

Soforum, *Web Forum* **Dec 2024 - Jan 2025**

- Designed and built a full-stack web forum from scratch, supporting user discussions, thread management, and secure authentication with a clean MVC architecture.
- Developed a modern, responsive frontend using React and Material UI, leveraging Next.js for server-side rendering and Redux for predictable client-side state management.
- Implemented a RESTful backend in Go to handle user and thread operations, incorporating JWT-based authentication middleware to enforce secure access control.
- Integrated PostgreSQL for persistent data storage and deployed the application to a cloud environment using Render, configuring the system for production use.